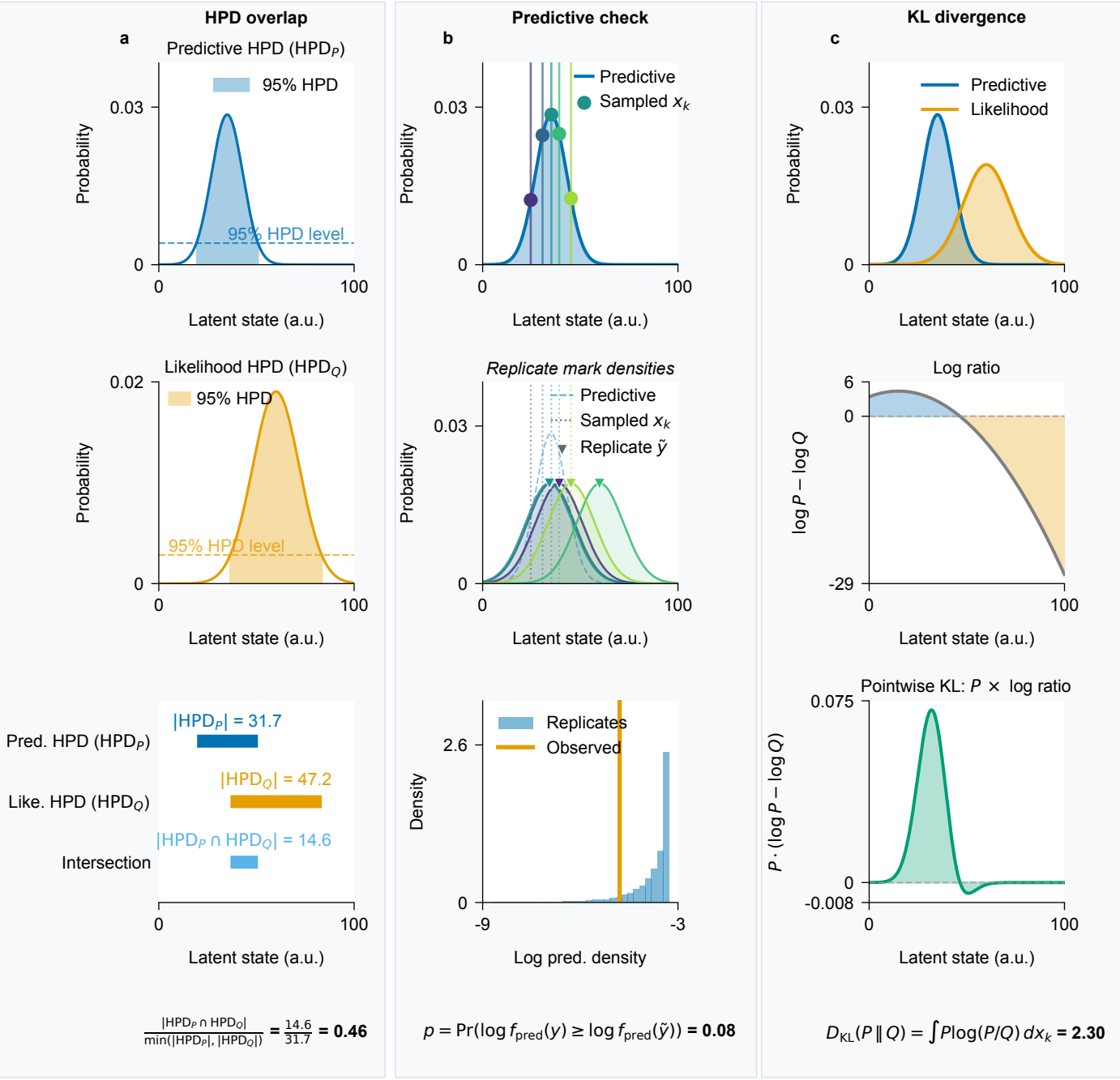

HPD overlap
a
Predictive HPD (HPD$_P$)
95% HPD
95% HPD level
Probability
0.03
0
0
100
Latent state (a.u.)
Likelihood HPD (HPD$_Q$)
0.02
95% HPD
95% HPD level
Probability
0
0
100
Latent state (a.u.)
|HPD$_P$| = 31.7
Pred. HPD (HPD$_P$)
|HPD$_Q$| = 47.2
Like. HPD (HPD$_Q$)
|HPD$_P$ ∩ HPD$_Q$| = 14.6
Intersection
0
100
Latent state (a.u.)
$\frac{|\mathrm{HPD}_P \cap \mathrm{HPD}_Q|}{\min(|\mathrm{HPD}_P|, |\mathrm{HPD}_Q|)} = \frac{14.6}{31.7} = 0.46$
Predictive check
b
Predictive
Sampled x_k
Probability
0.03
0
0
100
Latent state (a.u.)
Replicate mark densities
Predictive
Sampled x_k
Replicate $\tilde{y}$
Probability
0.03
0
0
100
Latent state (a.u.)
Replicates
Observed
Density
2.6
0
-9
-3
Log pred. density
$p = \Pr(\log f_{\mathrm{pred}}(y) \geq \log f_{\mathrm{pred}}(\tilde{y})) = 0.08$
KL divergence
c
Predictive
Likelihood
Probability
0.03
0
0
100
Latent state (a.u.)
Log ratio
6
0
$\log P - \log Q$
-29
0
100
Latent state (a.u.)
Pointwise KL: $P \times$ log ratio
0.075
$P \cdot (\log P - \log Q)$
0
-0.008
0
100
Latent state (a.u.)
$D_{\mathrm{KL}}(P \| Q) = \int P \log(P/Q)\, dx_k = 2.30$